



Voting systems

Task A: Single Transferable Vote (STV)

1) To decide the quota required to be elected, the following formula is used:

$$\frac{\text{total votes}}{\text{total seats} + 1} + 1$$

Calculate the quota for the following elections:

- a) 12 574 votes cast and three seats to be filled.
- b) 12 574 votes cast and four seats to be filled.
- c) 12 574 votes cast and five seats to be filled.

2) What do you notice about the number of votes required?

3) 20 000 votes are cast in an election. There are three seats available. Here is a table showing the breakdown of the votes.

First vote	Second vote			
A: 5813	B: 462	C: 1068	D: 1767	E: 2516
B: 7820	A: 1564	C: 2737	D: 539	E: 2980
C: 1292	A: 259	B: 543	D: 87	E: 403
D: 3983	A: 112	B: 92	C: 2350	E: 1429
E: 1092	A: 415	B: 317	C: 268	D: 92

- a) Calculate the quota required to win a seat.
$$\frac{\text{total votes}}{\text{total seats} + 1} + 1$$
- b) Based on the first vote, who has already won a seat?
- c) Assign all excess votes from any successful candidate(s) to the candidate(s) who were the most popular second vote.
- d) Who has won the final seat?

Task B: Alternative Vote (AV)

1) 20 000 votes are cast in an election. Here is a table showing the breakdown of the votes.

First vote	Second vote			
A: 5813	B: 462	C: 1068	D: 1767	E: 2516
B: 7820	A: 1564	C: 2737	D: 539	E: 2980
C: 1292	A: 259	B: 543	D: 87	E: 403
D: 3983	A: 112	B: 92	C: 2350	E: 1429
E: 1092	A: 415	B: 317	C: 268	D: 92

- a) How many votes are required to win the seat?
- b) Since no one has over half the vote, which candidate is eliminated?
- c) Assign the votes accordingly and produce an updated table.
- d) Which candidate(s) can win the election? Justify your answer.

